<Security Level 2>

IEGP

ENGINEERING DESIGN BASIS ELECTRICAL

DOC. NO. : IEGP-6351-8531-PH-000-1000

PROJECT NAME: PMC SERVICES FOR ETHYLENE GLYCOL PROJECT AT PARADIP REFINERY

TOYO JOB NO. : 6351

IndianOil

OWNER: INDIAN OIL CORPORATION LIMITED.

TOYO ENGINEERING

PMC : TOYO ENGINEERING INDIA PRIVATE LIMITED.

T1	05.02.2018	ISSUED FOR TENDER	SVP	CPM	RPT
A2	17.11.2017	ISSUED FOR CLIENT APPROVAL	SVP	CPM	RPT
A1	14.10.2016	ISSUED FOR CLIENT APPROVAL	SVP	CPM	RPT
O1	21.09.2016	ISSUED FOR CLIENTCOMMENTS	SVP	CPM	RPT
REV	**DATE**	**DESCRIPTION**	**MADE BY**	**REVIEWED BY**	**APPROVED BY**

TOYO ENGINEERING

TOYO ENGINEERING INDIA PRIVATE LIMITED

MUMBAI, INDIA

REVISION HISTORY	
REV.	DESCRIPTION AND/OR REVISED SHEETS
A1	Revised as per Client Comments
A2	Revised as per Client Comments and issued for Approval
T1	Issued for Tender

3.0 SITE CONDITIONS

A. Elevation above Mean Sea Level	:	Minimum 3.91m above Indian mean sea level (ISML) which corresponds to the site recorded flood level.
B. Minimum Temperature(T min)	:	11.3 Deg.C
C. Maximum Temperature(T max)	:	42.4 Deg.C
D. Design Temperature (T Design)	:	45 Deg.C
E. Relative Humidity at T max	:	99.7 %
F. Seismic Zone	:	Zone-3

4.0 SYSTEM DESIGN BASIS

4.1 Area classification and equipment requirements

4.1.1 All the areas within the battery limits shall be classified for the degree and extent of hazard from flammable materials as applicable. Classification of Hazardous areas for all locations shall be done in accordance with the PDRP-8531-SP-0001.

4.1.2 The electrical equipment installed in a hazardous area shall be designed to operate with a maximum surface temperature not greater than the maximum temperature given in the table below.

MARK OF CLASS T^O ON EQUIPMENT	LIMIT T^O OPERATION (^OC)	CLASS T^O OF EXPLOSIVE GAS ATMOSPHERE IN WHICH THE EQUIPMENT CAN OPERATE
T1	450°	T1
T2	300°	T1-T2
T3	200°	T1-T2-T3
T4	135°	T1 to T4
T5	100°	T1 to T5
T6	85°	T1 to T6

5.0 VOLTAGE LEVEL:

5.1 Terms used to define different voltage levels shall be as below:

HV – greater than 650V.

LV – less than or equal to 650V.

5.2 Incoming supply: 66kV, 3Phase, 50Hz, 40kA for 1 sec.

5.3 Utilization voltages

Service	Power Range kW	Voltage Volts	Phase	Freq Hz	Notes
Electric Motors	>4 MW	11kV ±6%(AC)	3	50±3%	To be confirmed by System Study.
Electric Motors	>160 ≤4 MW	6600V ±6% (AC)	3	50±3%	To be confirmed by System Study.
Electric Motors	0.18 – 160 kW	415V ±6% (AC)	3	50±3%	
Electric Motors	< 0.18 kW	240V ±6% (AC)	1	50±3%	
Lighting& Small Power	-	240V ±6% (AC)	1	50±3%	
Critical Lighting		220V (DC)			Through separate centralized Battery System
Instruments	-	110V (AC)	1	50±3%	Through Dedicated UPS
Switchgear Control		110V (DC)			Through Substation DC System
Telecomm/IT Security Systems/CCTV		240±6% (AC	1	50±3%	Through Dedicated UPS

6.0 SUB-STATION DESIGN:

6.1 The sub-station shall be located in a safe-area close to the load center. Consideration shall be given to vehicular traffic or any other factor that might affect the operation of the sub-station. Substation building shall be without any columns within switchgear room to ensure optimum space utilization.

6.2 Sub-station floor shall be raised above grade level and the space below the switchgear room shall be utilized as cable cellar. The cable cellar floor shall be 300 mm above finish ground level and shall be paved and cemented. Cable cellar shall not be pressurized. It shall be well ventilated with exhaust fans.

6.3 Battery banks shall be located in a separate adequately ventilated room in the substation buildings, along with the necessary exhaust system and water connection with sink. Floor of the battery room and walls up to 1.0 m height shall have acid/alkaline resistant protective material coating/tiling. Light fittings in this room shall be chemical resistant type. All electrical equipment installed in the battery room shall be flameproof type (of Gas group IIC) if there is possibility of concentration of Hydrogen / Hazardous gases greater than the prescribed limit as per relevant standards.

6.4 The HV capacitors and reactors for power factor improvement shall be located outside substation similar to transformers. However, their respective power & control panel shall be located suitably inside the substation.

6.5 The main entry for operating personnel in substation shall be through double door arrangement and with air lock lobby.

6.6 A separate entry of 3.0 m with rolling shutter shall be provided for drawing in all equipment for erection.

6.7 Equipment like oil filled transformers; neutral grounding resistors etc. shall be located in bays adjacent to the sub-station building. All bays shall have well drained floor, surfaced with gravel or other suitable material. Transformer bays shall have removable corrugated sheets at a height so as to facilitate removal of transformer core, radiators, conservator, top cover etc. for maintenance.. All bays shall have MS gates, main gate for removing the transformer and wicket gate for person entry. The gates shall be lockable. Transformers shall be located deep inside the bay to avoid rain (fine spray) falling on the transformer parts. Suitable rails shall be provided in the Power transformer bay for ease of maintenance. Partition walls between transformer bays and the ends of the last bay shall be of fireproof type and shall extend at least 600 mm above the height of the equipment. Height of the bay shall be decided so as to facilitate maintenance and easy removal of equipment and requirements of natural ventilation.

6.8 Oil immersed transformers with oil capacity exceeding 2000 liters, shall be provided with a soak pit of sufficient capacity to take the whole of the oil of the equipment. Where oil capacity of transformers exceeds 9000 liters, provision shall be made to drain away the oil to a separate waste oil tank/collection pit located away, through suitable drain pipes of 150 mm or 200 mm in diameter.

6.9 The Emergency Stop Push Button (NO contact for tripping & NC contact for close permissive) with suitable water protection canopy shall be placed inside the Transformer yard with easy access from outside the yard gate.

6.10 Layout of substation shall comply with OISD regulations, IE rules and any other statutory rules in general.

6.11 Substation shall have minimum entries as indicated below

- One entry for equipment, one normal entry for personnel on both sides
- Two emergency exits shall be provided in the middle on either sides for substation.

The Substation shall have all exit doors opening outwards.

7.0 EQUIPMENT SPECIFICATION:

7.1 General

7.1.1 The equipment shall in general conform to the latest revision of Indian Standards, IEC/Other International Standards applicable for the country of origin of the equipment.

7.1.2 Special tools/tackles required for operation & maintenance of equipments shall be supplied.

7.1.3 All Electrical equipment shall be tested/inspected with verification of complete compliance with approved vendor drawings.

7.1.4 For detail technical requirements of electrical equipment, Electrical Standard Doc. No. PDRP-8531-SP-001, Technical Specifications of Individual Equipment etc. enclosed elsewhere in the Tender document shall be referred. All electrical equipment supplied shall conform to technical requirements specified in the tender documents..

7.2 Power Management System (PMS):

7.2.1 Centralized supervisory Power Management system system shall be provided to permit remote monitoring of breaker operation and remote metering and same shall be hooked upto existing system in PP plant substation. Refer Technical Specification of PMS system enclosed elsewhere in Tender document for details.

7.2.2 MEG/ERU/O&U plant PMS system shall be fed from 110V AC Plant UPS system located in SRR. Similarly Cooling tower & BOOT- 3 substation PMS shall be fed from 110V AC Plant UPS located in Cooling Tower and BOOT-3 SRR respectively.

7.3 UPS and Battery charger & Battery

7.3.1 110V DC centralized battery charger shall be provided for control supply to HV & LV switchboards (PMCC & MCC) located in MEG/ERU/O&U substation and Cooling tower substation/BOOT-3 substation. Plant DC critical lighting shall be fed from separate 220V DC charger. Backup time of 60 minutes shall be provided.

7.3.2 240V AC UPS shall be provided in MEG/ERU/O&U substation for Telecomm/IT Security Systems/CCTV in MEG/ERU/O&U Substation and Cooling tower substation.

7.4 Fire Alarm system :

Fire alarm & detection system shall be provided for building areas & Plant areas of new MEG/ERU/O&U plant and same shall be hooked up with existing system in PP plant SRR. For further technical requirements regarding Fire alarm system, refer Technical specification for fire alarm system enclosed elsewhere in the tender document.

7.5 Communication system:

Paging & Telephone system shall be provided for new MEG/ERU/O&U plant and same shall be hooked up with existing PP plant Substation. For further technical requirements & make regarding

Communication system, refer Technical specification for these systems enclosed elsewhere in the tender document.

8.0 CABLE INSTALLATION:

Following philosophy should be used for cable installation

- Inside battery limit- All Power & control cable shall be laid inside RCC cable trenches. For loads located at elevation, cables shall be laid in cable trays/ conduits as required.
- Outside battery limit: All Power & control Cables shall be laid on cable trays mounted on pipe racks or inside RCC cable trenches.
- The Lighting cables in ISBL/OSBL area shall be laid in overhead trays/conduits.
- The Communication cables in ISBL/OSBL area shall be laid in overhead Instrumentation cable trays/conduits/ducts
- Separate trays shall be provided for HV/LV power, control and Telecommunication system cables.

9.0 LOAD SHEDDING:

Contingency based Load shedding with back-up under frequency based load shedding scheme shall be provided.

Load shedding scheme shall include HV & LV Breaker controlled feeders, Contactor controlled LV motor feeders upto 37kW. Motors rated less than 37kW shall be excluded from load shedding scheme.

All feeders identified for load shedding shall be provided with proper hardware (CTs, Open bus protocol in numerical relays, transducers etc.)

Critical loads like UPS, battery chargers, critical process units and equipment shall not be covered under load shedding.

10.0 GUARANTEE FOR ELECTRICAL EQUIPMENTS:

10.1 The performance figures which will be indicated by vendor shall be guaranteed subject to the tolerance permitted by standards.

10.2 If any equipment supplied by vendor fails at site during erection, commissioning or service (within guarantee period), the vendor shall repair and put back into successful operation the failed equipment within the time frame and procedure of repair agreed with the purchaser depending on nature of failure at no extra cost to Owner.

10.3 OWNER/PMC reserves the right to reject the equipment in case of failure to meet the guaranteed electrical

parameters. However, the vendor will be given an opportunity to rectify his equipment at his own cost. Also Owner reserves the right to use rejected equipment till it is rectified.

11.0 TRAINING:

Training shall be provided for 2 nos. Owner engineers at manufacturer's works for period of one week for the following equipment as a minimum being supplied by Contractor. All expenditure in connection with travel and stay shall be borne by Owner for their engineers.

1) GIS.
2) Power Management system (PMS).
3) UPS.
4) HV VFD.
5) Numerical Protection Relays
6) Paging system
7) Fire alarm system

IndianOil	PARADIP REFINERY PROJECT PROJECT SPECIFICATION	PDRP-8531-SP-0001 PAGE : 1 of 33 REV : F3
	ELECTRICAL STANDARD	

Client's Name:	Indian Oil Corporation Limited (IOCL)
Project Title:	Paradip Refinery Project (PDRP)
Project Location:	Paradip, Orissa State, India
Document Category	Class 1

REVISION	F3	SIGNATURE
DATE	30 Nov 10	
ORIG BY	Paul Perry	P. Perry.
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REVISION HISTORY

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O1	12.04.07	First Issue for Comment	A K Bhan
A1	19.06.07	Issue for Design	C.Oyarzábal
A2	06.09.07	Approved for Design	M Alam
A3	12.02.08	Revised and approved for Design	M Alam
F1	12.08.09	For Project Implementation Phase	P.Perry
F2	15.10.09	For Project Implementation Phase	J. Carver
F3	30.11.10	Final with ATI's Incorporated	P.Perry
F4	07.11.14	Updated by TKIS India for Tender	M. Deshmukh

Changes to this document are highlighted as per this statement and are summarised below:

F5	11.12.14	Issued for Tender after incorporating IOCL comments	M.Deshmukh

Section 4 referenced document number updated
6.1 Para 1, reference clarified
6.1 para 9, aligned with design note 3210-8820-TN-0036
6.2 referenced document number updated
7.2 Table, added ref to notes for clarity
7.2 note 7, EPCM-2-MPMC-TQR-0344
7.2 note 9, referenced document number updated
8.1 para 1, reference to PCC and PMCC added for clarity
8.1 page 13, added rating for PCCs and MCCs as per 3210-8820-TN-0036 page 10 & 11
8.1 page 13, added 'e.g. redundant unit and manual bypass' as 3210-8820-TN-0036, page 35
8.1 page 13, added ref to 'other critical loads…' as 3210-8820-TN-0036, page 35
9.2 table page 45 aligned with 3210-8820-TN-0036 page 35
11.2 aligned

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Prepared by : HGG	Checked by : MSD	Approved by : RPM/HMC

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F7	05.02.2018	ISSUED FOR TENDER	SVP	CPM	RPT

1.0 SCOPE

This Standard gives the basic requirements for the design, equipment and materials of the Electrical Installation for the Paradip Refinery Project and its associated facilities. This Standard should be supplemented, as applicable, by other project Standards.

2.0 ABBREVIATIONS

The following abbreviations are used in the Electrical Standard

AC	Alternating Current
ACB	Air Circuit Breaker
ANAN	Air Natural Air Natural
ASB	Auxiliary Supply Boards
BIS	Bureau of Indian Standard
CB	Circuit Breaker
CCE	Chief Controller of Explosives
CMRI	Central Mining Research Institute
CPRI	Central Power Research Institute
DC	Direct Current
DCS	Distributed Control System
DGMS	Directorate General of Mining Services
EMMS	Electrical Maintenance and Monitoring System
ERTL	Electronic Regional Test Laboratory
ETP	Effluent Treatment Plan
FA	Fire Alarm
HMI	Human Machine Interface
HPS	High Pressure Sodium
HV	High Voltage (>1 kV)
I.E	Indian Electricity
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IOCL	Indian Oil Corporation Limited
ISBL	Inside battery Limit
LDB	Lighting Distribution Boards
LED	Light Emitting Diodes
LV	Low Voltage (Š1 kV)

MCB	Miniature Circuit Breaker
MCC	Motor Control Centre
NER	Neutral Earthing Resistor
ONAN	Oil Natural Air Natural
PA	Public Address
PCC	Power Control Centre
PMCC	Power & Motor Control Centre
PT	Potential Transformer
PVC	Poly Vinyl Chloride
SFU	Switch Fuse Unit
SLD	Single Line Diagram
UL	Underwriters Laboratory
UPS	Uninterrupted Power system
VSD	Variable Speed Drives

3.1 BASIS OF DESIGN

The Electrical Installation shall provide: -

- Safety to personnel.
- Reliability of service
- Provision for future requirements if specified.
- A graded system of protective devices.
- Equipment with adequate:
  - Insulation levels.
  - Interrupting capacity.
  - Make Duty capability.
  - Through Fault capability.
- Continuous current carrying capacity.
- Control and indication.
- Facilities for operation and maintenance.
  - Maximum interchangeability of equipments/spares.
  - Aesthetics
  - Minimal fire risk

The design shall be assessed with regards to health and safety during construction, commissioning, operation and maintenance. Where practical, potential risks shall be mitigated at source and incorporated into the design.

4.1 CODES AND REGULATIONS

The assemblies and equipment therein shall conform to the requirements of the latest revisions and amendments of the applicable IEC standards, local laws, codes and regulations applicable to India, as indicated in documents PDRP-8820-IN-0002 and PDRP-8820-IN-0003, including the following specific standards:

- BS 7671 Requirements for Electrical Installations (I.E.E. Wiring Regulations).
- The Institute of Petroleum Model Code of Safe Practice, Part 15, Area Classification Code for Petroleum Installations.
- The Institute of Petroleum Model Code of Safe Practice, Part 1, Electrical Safety Code.
- Electricity Association G5/4 Engineering Recommendation.
- EMMUA Publication No 133 Specification for Underground Armoured Cables Protected against Solvent Penetration and Corrosive Attack.
- Electrical Research Association Report No. 69-30.
- British, European and IEC Standards.
- Local Laws and Regulations.
- The main local codes, standards & statutory regulations (latest version) of the following shall be applicable:
  - Indian Electricity act & rules framed there under.
  - The factory act & regulations.
  - The Petroleum rules.
  - Fire Insurance Regulations.
  - National Electrical Code (NEC)-SP-30 –BIS Publication.
  - OISD Standards.

OISD-STD-110	Recommended Practices On Static Electricity
OISD-STD-113	Classification Of Area For Electrical Installation at Hydrocarbon & Handling Facilities
OISD-STD-147	Inspection & Safe Practices During Electrical Installation
OISD-STD-149	Design Aspects For Safety In Electrical Systems
OISD-STD- 173	Fire Protection System For Electrical Installations
OISD-STD-180	Lightning Protection

- Requirements of other Indian statutory bodies as applicable;

  Central Electricity Authority (CEA), State Electrical Inspectorate, Chief Controller of Explosives (CCE), Directorate General of Mining Services (DGMS), India.

5.1 EQUIPMENT AND MATERIALS

- Equipment and materials shall be provided which conform to British Standards, European or IEC Standards and Indian standards as applicable.
- Equipment and Materials, for use in a hazardous area shall be certified by an ATEX Notified Body as complying product directive 94/9/EC (ATEX 95) and/or have DGMS/CCE approval. The Vendor shall provide a Certificate of Conformity.

6.1 AREA CLASSIFICATION

6.2 General

All the areas within the battery limits shall be classified for the degree and the extent of hazard from flammable materials. The Area Classification shall be based on PDRP-8820-SP-0001 and IS5572.

Only those potential sources of hazard within the IOCL Scope of Work will be considered unless IOCL is informed of any other relevant source of hazard.

Following factor shall be considered for proper selection of electrical equipment for use in hazardous area:

- Area classification i.e. Zone.
- Gas classification i.e. Group – The characteristic of the gas or vapour involved in relation to the ignition current or minimum ignition energy and safe gap data.
- Dust classification i.e. Group –The characteristic of the dust involved.
- Temperature classification – The ignition temperature of the gas, vapour or dust involved or the lowest value of the ignition temperature if more than one combustible material is present.
- Environmental condition in which apparatus is to be installed. The selected electrical apparatus shall be adequately protected against corrosive and solvent agencies, water ingress, thermal and mechanical stresses as determined by the environmental conditions.

All electrical equipment installed in hazardous areas shall be selected as per latest standard IEC 60079-14 Ed 4.06:2007 to be adopted for "Explosive Atmosphere – Part 14 Electrical Installations design, Selection & Erection", IS -5571, Guide for Selection of Electrical Equipment for Hazardous Area and IS-14154, Electrical Enclosures for Combustible Installations and shall meet

the requirements of relevant IS and Statutory regulations. However, electrical equipments for Zone 2 areas as a minimum shall be Ex e/Ex n type as per IS/IEC codes. Increased safety, Ex e equipment shall not be used in Zone 1 areas. Industrial electrical equipment (even though permitted for use in Div.2 area as per NFPA-70) shall not be used in Zone 2 areas. Type of explosion protection to be used for individual equipment shall be as specified in the data sheets.

For pressurised electrical equipment Ex p on failure of pressurisation system, the following minimum actions shall be provided:

Area classification	Minimum Action to be Taken
Zone 1	Alarm and Switch off
Zone 2	Alarm

Electrical equipment for hazardous areas shall be certified by CMRI/CPRI/ERTL or equivalent recognized independent test house such as BASEEFA / LCIE /PTB /UL /FM. All equipment (indigenous and imported) shall also have valid statutory approvals as applicable for use in the specified hazardous area. The indigenous equipment shall conform to Indian standards and shall be approved by CCE/DGMS as applicable. All indigenous flameproof equipments shall have BIS license and marking as required by statutory authorities.

The hazardous area motors (Ex e or Ex d) fed by variable frequency drives shall be certified as a combination for the specified location by Independent test house and shall have valid statutory approvals. Non-sparking (Ex n) motors shall not be used with frequency converters/Variable frequency drives. In case of Ex d flameproof motors, the motor and variable frequency drive combination may not need to be certified together if direct temperature control by embedded temperature sensors are provided, which will disconnect the motor.

6.3 Zones

Unit's areas that have potentially explosive gas atmospheres shall be classified in accordance with what is indicated in document PDRP-8820-SP-0001 as follows:-

Zone 0 — in which an explosive gas-air mixture is continuously present, or present for long periods.

Zone 1 — in which an explosive gas-air mixture is likely to occur in normal operation.

Zone 2 — in which an explosive gas-air mixture is not likely to occur, and if it occurs it will only exist for a short time.

Units areas that have a combustible dust hazard shall be classified in accordance with what is indicated in document 3210-8820-SP-0001 as follows:-

Zone 20 Area in which an explosive atmosphere in the form of a cloud of combustible dust in air is present continuously, or for long periods or frequently.

Zone 21 Area in which an explosive atmosphere in the form of a cloud of combustible dust in air is likely to occur in normal operation occasionally.

Zone 22 Area in which an explosive atmosphere in the form of a cloud of combustible dust in air is not likely to occur in normal operation but if it does occur, will persist for a short period only.

Areas of unit not classified as one of the above shall be designated as being non-hazardous.

7.1 ELECTRICAL EQUIPMENT ENCLOSURES

7.2 Enclosure

IP ratings to IEC 60529 or equivalent Indian standards shall be applied.

7.3 Hazardous Areas

For gas and vapour risk areas, electrical equipment and the types of protection for enclosures shall comply with IEC 60079 / IS13346 and shall meet requirements of statutory regulations.

The requirements for construction of electrical apparatus for use in potentially explosive atmospheres shall comply with relevant part of IEC 60079 or equivalent Indian Standards.

Electrical Equipment for dust risk areas shall comply with IEC 61241-10 and IS 14154-1, IS 14154-2 and IS 15142.

The type of protection to be applied to the electrical apparatus relative to the area classification zone shall be as follows:

	TYPE OF PROTECTION								
ZONE	ia	ib	d	e	p	n	o	q	DEGREE OF PROTECTION OF ENCLOSURE
0	X								Min. IP55
1	X	X	X	*	X				Min. IP55
2	X	X	X	X	X	X	X	X	Min. IP55

* Increased safety enclosure is not permitted in Zone 1 area. Industrial equipment (even though permitted for use in Div. 2 area as per NFPA-70) shall not be used in Zone 2 areas.

Types of protections 's' and 'm' may be considered dependent on the application and on appropriate certification being available.

The maximum surface temperature shall not exceed the ignition temperature of the gases, dusts or vapours involved. This shall be indicated by the T Class (T1-T6) of the apparatus as defined in IEC 60079 / IS 8239.

The electrical equipment for hazardous areas shall be selected as per IS 5571. The guide providing minimum requirement is summarized below:

Equipment		Zone 1		Zone 2	
		Gas Group	Gas Group	Gas Group	Gas Group
		IIA, IIB	IIC	IIA, II B	IIC
LV Motors (Š160kW) See Note 7 & 9		Ex d	Ex d	Ex n Note 3&7	Ex e Note 3,7&10
HV Motors (>160kW) See Note 7 & 9		Ex d/Ex p	Ex d See Note 1	Ex e/ Ex p Note 1,10,11	Ex e/Ex p Note 1,10,11
Push Button station		Ex d	Ex d	Ex d	Ex d
Plug & Socket		Ex d	Ex d	Ex d	Ex d
Welding Receptacle		Ex d	Ex d	Ex d	Ex d
Lighting Fixtures					
i	Lighting fitting	Ex d	Ex d	Ex e/Ex n Note 3	Ex e/Ex n See Note 8
ii	Control gear Box	Ex d	Ex d	Ex d	Ex d
Junction Boxes		Ex d	Ex d	Ex e	Ex e
Hand lamps					
i	Light fitting	Ex d	Ex d	Ex d	Ex d
ii	Transfer unit	Ex d	Ex d	Ex d	Ex d
iii	Plug & Socket	Ex d	Ex d	Ex d	Ex d
Break glass Unit (Fire Alarm System)		Ex d	Ex d	Ex d	Ex d
Lighting Panel/power Panel		Ex d	Ex d	Ex d	Ex d
Transformer		Hermetically Sealed with Surface temperature Not exceeding 200^{0}C			

Notes:

1. Ex p could be used in special cases with the approval of IOCL.
2. The electrical equipment for hazardous areas shall generally be suitable for gas group IIA/IIB/IIC and temperature classification as applicable to the selection type of explosion protection. In case of hydrogen or hydrocarbon mixture having more than 30% hydrogen, the gas group to be considered shall be IIC.
3. As additional safety features, the following requirement for electrical equipment shall be followed;
   - All electric motors for agitators/mixers handling flammable materiel shall be flameproof type (Ex d) irrespective of the area being classified as zone 2 or zone 1.
   - All electric motors for vertical oil sump pumps including ETP area shall be flameproof type (Ex d).
   - Irrespective of the area classification (whether zone 1 or zone 2), all lighting fixture within the storage areas shall be flameproof type (Ex d).
   - All emergency/ critical lighting fixtures and associated junction boxes in hazardous areas (whether zone 1 & zone 2) shall be flameproof type (Ex d).
4. Even though fired heaters in process units are not considered for area classification, all electrical equipments associated with fired heaters in process units shall as a minimum be suitable for installation in zone 2 area.
5. Where air conditioning system is designed considering ammonia as refrigerant, the room housing air conditioning equipment shall be adequately ventilated to classify it as safe area. For additional safety the following shall be considered:
   - 100% stand by system for ventilation.
   - Location of MCC /local panels in adjacent separate room.
   - Instrumentation to be flameproof type or hermetically sealed.
   - AC unit room motors with type 'e' protection.
   - Lighting in AC unit room suitable for zone 2 area.
6. Compressor sheds in hazardous area shall be designed to allow adequate ventilation to allow area classification as Zone 2 Lighting equipment, EOT crane etc. in the shed shall be flameproof type. All other electrical equipment shall be suitable for zone 1 or zone 2 area depending on extent of hazard.
7. Motors fed from variable frequency drive in hazardous area application shall be type tested as unit with the VSD panel in case of Ex e type of motors and provision for tripping the motor in case temperature rise

exceeds the permissible limits, shall be considered for flameproof motors as per relevant standards. VSD motors in Zone 2 areas shall be Ex d.

8. Ex e Fluorescents, Ex n other type.

9. Refer to Electrical System Specification PDRP-8531-SP-0009 for utilization voltage level of motors.

10. For Ex e motors, if the sum of risk factors is greater than 6 as per Section 5.2.4.3 and Table 4 of IEC 60079-7, the machine or representative sample shall be tested in accordance with Section 6.2.3.2 of IEC 60079-7, or the machine shall be constructed to allow special measures to be employed to ensure that its enclosure does not contain an explosive gas atmosphere at the time of starting. The machine instructions provided, in accordance with IEC 60079-0, shall include on the implementation, where required, of additional measures.

11. For Ex e motors rated >1kV, Section 5.2.7 of IEC 60079-7, type tests in accordance with Section 6.2.3.1 shall be conducted, and

- the machine shall be fitted with anti condensation heaters, and
- the machine shall be constructed to allow additional measures to be employed to ensure that its enclosure does not contain an explosive gas atmosphere at the time of starting. The machine instructions provided, in accordance with IEC 60079-0, shall include on the implementation, where required, of additional measures.

7.4 Non-Hazardous Area

7.3.1 Indoor Locations

Electrical equipment in a relatively dust free environment (e.g. Substation Building) shall be standard industrial equipment. Switchboards shall have a minimum degree of protection of IP 42. Ancillary apparatus shall have a minimum degree of protection IP 42. Electrical apparatus in dusty areas shall have a minimum protection of IP55 and with appropriate protection class for classified areas.

The substation floor shall not be considered as part of the enclosure.

7.3.2 Outdoor Locations:

The minimum enclosure for electrical equipment shall be IP 55. Electrical equipment shall be suitable for the environmental conditions (e.g. dust, moisture, snow or rain) with additional protection provided (e.g. rain shields, sun shields) where required.

8.1 DISTRIBUTION SYSTEM DESIGN

8.2 General

Distribution systems shall be of the secondary selective type, unless the unit is of a type or in a location where single radial or spot networks are standard